# SHARK TANK STARTUP INVESTMENT ANALYSIS

Data Analyst Internship Project Report

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**Program:** B.E. Computer Science Engineering – Data Science

**Institution:** Lords Institute of Engineering & Technology

**Tools:** Python, Pandas, Excel and Microsoft Power BI

## 1. Introduction

Startup investment analysis helps identify funding patterns, industry preferences, investor participation and deal outcomes. Shark Tank provides a useful dataset for understanding how startups present funding requirements and how investors respond. This project analyzes Shark Tank India startup data to convert raw investment records into meaningful business insights through data cleaning, exploratory analysis and interactive visualization.

## 2. Abstract

The objective of this project is to analyze startup investment trends using a Shark Tank dataset containing information about startups, industries, funding requests, actual deal amounts, equity and participating Sharks. The dataset was cleaned and analyzed using Python and Pandas. Key metrics and summary tables were exported and used to create dashboards in Excel and Power BI. The final dashboard presents startup counts, total investment, accepted deals, average deal amount, acceptance rate, industry-wise investment, Shark-wise investment, top-funded startups and comparison of requested versus actual funding.

## 3. Tools Used

• Python
• Pandas and NumPy
• Matplotlib
• Jupyter Notebook
• Microsoft Excel
• Microsoft Power BI

## 4. Steps Involved

**Dataset Collection:** A Shark Tank India dataset containing startup details, industries, funding requests, deal information and investor participation was selected.

**Data Cleaning:** The dataset was inspected for dimensions, missing values, data types and duplicate records. Date fields were converted, missing values were handled, and duplicate rows were removed.

**Exploratory Analysis:** Industry-wise investment, Shark-wise investment, deal counts, acceptance rate, average deal values, top-funded startups and differences between requested and actual funding were calculated.

**Dashboard Development:** Analytical summaries were exported as CSV files and used to create dashboards in Excel and Power BI.

## 5. Key Results and Findings

| Metric | Result |
|---|---|
| Total startups analyzed | 290 |
| Accepted deals | 175 |
| Not accepted deals | 115 |
| Total investment recorded | 10,945.6 |
| Average deal amount | 37.74 |
| Acceptance rate | 60.34% |
| Highest investment industry | Food |
| Highest cumulative investment by Shark | Aman |

The analysis indicates that Food had the highest total recorded investment among the industries in the dataset. Aman had the highest cumulative investment among the listed Sharks. The dataset recorded 175 accepted deals out of 290 startups, producing an acceptance rate of approximately 60.34%. The dashboard also compares requested amounts with actual investment received, helping illustrate negotiation outcomes.

## 6. Deliverables

- Cleaned Shark Tank dataset
- Python/Jupyter analysis notebook
- CSV analytical summary files
- Excel investment analysis dashboard
- Interactive Power BI dashboard (.pbix)
- Project documentation and findings report

## 7. Conclusion

This project successfully transformed raw Shark Tank startup investment data into a structured analytical solution. Python and Pandas were used for data preparation and analysis, while Excel and Power BI communicated findings through dashboards. The project demonstrates practical skills in data cleaning, exploratory data analysis, KPI creation, aggregation, visualization and business storytelling. The resulting dashboard provides a clear view of industry funding patterns, investor participation and startup deal outcomes.

**Note:** Investment values are reported in the same units as provided in the source dataset.